
Robust M3m3 H@te D3t3ct!on: Robustness of CLIP-fusion classifiers under naturalistic and adversarial perturbations

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Abstract

Hateful-meme detectors are deployed against users who actively obfuscate content, yet are almost always benchmarked only on clean inputs. We audit a CLIP-fusion classifier on the Hateful Memes dataset under *naturalistic* label-preserving edits (typos, leetspeak, censoring, compression, blur, crops, occlusions, and composites) and *white-box* L_∞ image attacks. The audit exposes two failures: the model is a text shortcut whose image branch is effectively dead (image-only AUROC 0.59 vs. a dedicated image baseline 0.63), so text edits dominate degradation; and it is undefended against white-box PGD (AUROC < 0.01 at $\epsilon=2/255$). A lightweight pair of training-time interventions — clean $\leftrightarrow$ perturbed KL-consistency plus per-example text-modality dropout — revives the image branch (image-only AUROC 0.59 $\rightarrow$ 0.66, exceeding the dedicated baseline) and raises the count of fully naturalistic-robust held-out examples by 76%, at *zero* clean-accuracy cost. The defense is class-asymmetric: 97% of the predictions it corrects are benign memes, acting as a false-positive veto. All results are reported over three seeds and three splits.

Keywords: hateful memes, multimodal robustness, CLIP, consistency regularization, modality dropout.

1. Introduction

Automated hateful-meme moderation is adversarial by construction: the users a detector should catch actively rewrite captions and re-encode images to evade it. Yet such models are reported almost exclusively on *clean* benchmark accuracy [1, 2], which says little about deployed behaviour once an everyday user adds a typo or a malicious actor crafts an imperceptible pixel perturbation. We therefore audit a CLIP-fusion classifier under two complementary

threat models — naturalistic, label-preserving obfuscations and white-box image attacks — and ask whether a cheap training-time intervention can close the naturalistic gap without sacrificing clean accuracy. Using a controlled perturbation suite (8 text $\times$ 11 image attacks $\times$ 3 severities, plus composites, deterministically seeded) over three seeds and three splits, we contribute: (i) an audit showing the model is a *text shortcut* with a dead image branch, so text edits dominate naturalistic degradation while white-box PGD breaks every recipe; (ii) a lightweight recipe (KL consistency plus text-modality dropout) that *revives* the image branch at zero clean-accuracy cost; and (iii) a failure analysis showing the defense chiefly prevents *false positives* on benign memes. We target the audit and recipe, not leaderboard accuracy.

2. Related Work

Hateful-meme detection is multimodal because harmful meaning often emerges only from the image–caption combination; the Hateful Memes Challenge enforces this with benign confounders that flip the label when a single modality changes, and reports a large model–human gap that survives careful dataset curation [1]. Recent systems exploit the shared CLIP embedding space [3]: Hate-CLIPper cross-correlates the image and text embeddings through a feature-interaction matrix [2], and MemeCLIP adds lightweight adapters and a cosine classifier for small, imbalanced meme sets [4]. These optimise clean benchmark accuracy and do not characterise behaviour under corruption or attack. A second line asks whether such models use both modalities: cross-domain performance is often text-driven while image features are dataset-sensitive [5]. Robustness to label-preserving corruptions and white-box L_∞ attacks [6, 7, 8] is well studied for unimodal vision but rarely measured for CLIP-fusion meme detectors. We close this gap with one controlled suite that diagnoses *which* modality is fragile and compares clean against robustness-aware training.

3. Method

Data. We use the Hateful Memes dataset [1]: 8,500 train / 500 dev memes, plus the labelled `test_seen` ($n=1000$)

and `test_unseen` ($n=2000$) splits for held-out evaluation, keeping the official split structure for comparability.

Classifier. A dual-encoder CLIP-fusion model on OpenCLIP ViT-B/32 maps the image and tokenized caption to $v, t \in \mathbb{R}^{512}$. Pixel normalization is a module *inside* the network, so the forward pass is differentiable w.r.t. raw $[0, 1]$ pixels (shared by white-box attacks and consistency training). A fusion head consumes

$$z = [t; v; |t - v|; t \odot v] \in \mathbb{R}^{2048} \quad (1)$$

($|t - v|$ captures cross-modal disagreement, $t \odot v$ alignment) through $\text{LayerNorm} \rightarrow \text{Lin}(2048, 512) \rightarrow \text{GELU} \rightarrow \text{Lin}(512, 1)$. We freeze the CLIP encoders and train only the 1.05M-parameter head (no gain to fine-tuning 150M params on 8.5k memes); zeroing either embedding before the head yields unimodal forward modes for ablation.

Perturbation suite. Eight text attacks (e.g. leetspeak, char deletion/swap, censoring, typos) and eleven image attacks (Gaussian noise, blur, JPEG compression, brightness/contrast, translation, crop, occlusion, typographic overlay) each have low/medium/high presets, with *medium* the realistic level reachable with an off-the-shelf editor and *high* a stress ceiling. Cells use SHA-256-keyed per-example seeds; *composite* cells apply $K \in \{2, 4\}$ perturbations per sample to model a determined user.

Robustness-aware training. Each batch pairs a clean view with an on-the-fly perturbed view (perturb text/image/both with prob. 0.5/0.3/0.2). **AugOnly** supervises both views, $\mathcal{L}_{\text{Aug}} = \text{BCE}(y, \ell_{\text{clean}}) + \alpha \text{BCE}(y, \ell_{\text{pert}})$. **KL** adds consistency to the detached clean prediction, $\mathcal{L}_{\text{KL}} = \mathcal{L}_{\text{Aug}} + \beta \text{KL}(p_{\text{clean}} \| p_{\text{pert}})$ with $p = \sigma(\ell)$, $\alpha=1$, $\beta=0.5$. **KLDrop** additionally zeros the text embedding for a fraction p of examples, forcing a real image-only prediction; we sweep $p \in \{0, 0.10, 0.15, 0.20, 0.25, 0.30, 0.50\}$. The clean baseline minimizes BCE alone, with AdamW, cosine decay, and early stopping on dev macro-F1.

White-box attacks. We probe the worst case with image-space FGSM [6] and PGD [7] (L_∞ , text fixed) over $\epsilon \in \{1, 2, 4, 8\}/255$. As the targeted remedy for this regime we implement Madry-style adversarial training and a TRADES-style [8] consistency variant; their full ϵ -curve evaluation is ongoing and out of scope here.

Metrics. We report AUROC (threshold-free, primary), macro-F1, and FPR/FNR (moderation has asymmetric costs). Per cell we report the robustness gap $\Delta\text{AUROC} = \text{AUROC}_{\text{clean}} - \text{AUROC}_{\text{att}}$ and attack success rate (ASR), with thresholds fixed on clean dev and reused under attack. Every recipe runs for 3 seeds (mean $\pm$ std).

Model	AUROC	F1	Acc.	Params
Majority (all-0)	0.500	0.333	0.500	0
Image-only (ded.)	0.628	0.574	0.574	1.05M
Text-only (ded.)	0.632	0.597	0.598	1.05M
Multimodal fusion	0.734	0.642	0.649	1.05M

Table 1. Clean dev results (3-seed mean, $\sigma \leq 0.007$). Fusion beats the best unimodal baseline.

4. Validation

We evaluate every recipe over 3 seeds on the test sets, against a majority baseline and dedicated unimodal classifiers; image and text-only heads trained from scratch.

Audit: a text shortcut with a dead image branch. Table 1 shows fusion helps (+0.10 AUROC over the best unimodal baseline), so the model is not a pure text classifier. Yet its image-only forward AUROC (0.59) is *below* the dedicated image baseline (0.63) while text-only matches its baseline: the decision is routed through text and the image branch left under-trained. This predicts text-edit fragility, and indeed the worst text cell costs $\Delta\text{AUROC} = 0.11$ vs. 0.04 for the worst image cell (Table 2) — the model is even *more* text-fragile than the dedicated text baseline. White-box attacks set the floor: PGD collapses AUROC from 0.73 to < 0.01 (ASR > 0.98) at $\epsilon=2/255$ and to 0 at $\epsilon \geq 4/255$, which augmentation cannot address, so we scope the defense to the naturalistic threat model.

Defense: reviving the image branch at zero clean cost.

The dropout sweep (Fig. 1b) shows a free-lunch window: image-only AUROC rises monotonically with p , peaking at 0.668 ($p=0.25$), while multimodal AUROC stays within seed noise through $p \leq 0.25$. We pick $p=0.15$ (KLDrop-p015) as default, $p=0.50$ as a robustness-heavy alternative. KLDrop *redistributes* capacity rather than adding signal (Fig. 1a): its image-only forward reaches the dedicated baseline on every split, text-only unchanged. Table 2 gives the held-out trade-off: KL preserves clean AUROC, KLDrop-p015 even nudges it up while improving every robustness column, and the count surviving all 57 naturalistic cells rises $222 \rightarrow 391$ (+76%). Composites are the real threat: two combined text edits cost the clean model $\Delta\text{AUROC} = 0.090$ (above the worst single cell), and KLDrop cuts medium-severity composite gaps by up to $\sim 50\%$ (e.g. $0.050 \rightarrow 0.024$).

Why it works: a class-asymmetric false-positive veto. A per-example analysis on `test_unseen` reveals the mechanism (Fig. 2): where KLDrop-p015 and KL disagree, 97% of KLDrop’s corrections are on *benign* memes (95% CI [94, 99], $n=131$) vs. 75% on hateful memes for KL. The revived image branch vetoes the text branch when an edited caption looks superficially hateful but the image does not support it — the over-moderation failure that erodes trust.

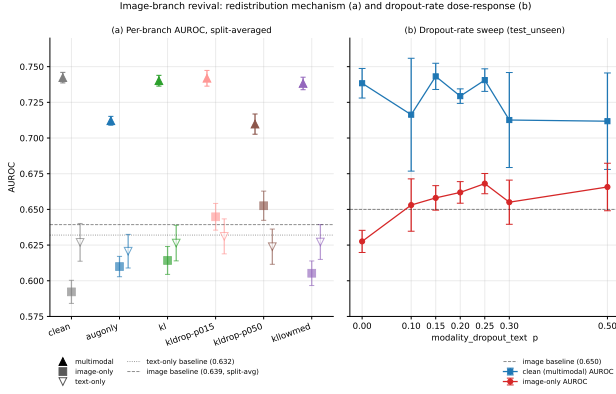


Figure 1. (a) Per-branch forward AUROC (mean over 3 splits $\times$ 3 seeds). On `clean` only the text branch carries signal; KLDrop-p015/p050 push image-only to or above the dedicated baseline (dashed) while text-only is unchanged. (b) Dropout sweep (`test_unseen`).

Recipe	Clean AUROC	W-cell text Δ	Image-only	Robust cnt/2000
clean	0.738	0.110	0.602	222
AugOnly	0.708	0.063	0.620	282
KL	0.738	0.065	0.628	352
KLDrop-p015	0.743	0.069	0.658	391
KLDrop-p050	0.712	0.049	0.666	331

Table 2. Recipes on held-out `test_unseen` (3-seed mean). KLDrop-p015 improves every robustness column while increasing clean AUROC.

The choice is thus deployment-conditional: false-positive-sensitive contexts prefer KLDrop, recall-sensitive ones prefer KL; the direction holds on all three splits.

Limitations..

5. Conclusion

A CLIP-fusion hateful-meme detector is a text shortcut with a dead image branch — fragile to the text edits a real user applies, and undefended against white-box PGD. Our recipe (KL consistency plus text-modality dropout) revives the image branch above a dedicated baseline and raises held-out naturalistic robustness by 76% at zero clean-accuracy cost, acting as a class-asymmetric false-positive veto for benign content; the white-box regime remains open for adversarial training. Hateful-meme detectors should thus be reported with a modality-resolved robustness audit, not clean accuracy alone, since the dominant failure — over-flagging benign speech after a trivial edit — is invisible on clean benchmarks. These are research prototypes, not moderators; we release the suite for defensive evaluation.

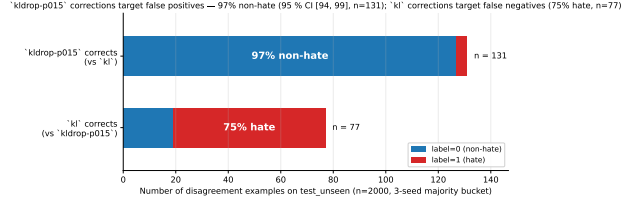


Figure 2. Per-example disagreements between KLDrop-p015 and KL on `test_unseen` ($n=2000$), split by gold label. KLDrop’s corrections are 97% benign (`label=0`) — a false-positive veto — while KL’s corrections are mostly hateful (`label=1`).

References

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